You Try/Practice:

For problems 1 - 12, solve for x.

1) $5x + 3 - 3x = -7$

$2x+3=-7$
$-3 \quad -3$
$\frac{2x}{2} = \frac{-10}{2}$
$x=-5$

2) $12x + 14 + 10x = 80$

$22x+14=80$
$-14 \quad -14$
$\frac{22x}{22} = \frac{66}{22}$
$x=3$

3) $11x - 9 - 7x = 15$

$4x-9=15$
$+9 \quad +9$
$\frac{4x}{4} = \frac{24}{4}$
$x=6$

4) $3 + 4(x + 5) = 31$

$3+4x+20=31$
$4x+23=31$
$-23 \quad -23$
$\frac{4x}{4} = \frac{8}{4}$
$x=2$

5) $27 = 3 - 3(6 - 2x)$

$27=3-18+6x$
$27=-15+6x$
$+15 \quad +15$
$\frac{42}{6} = \frac{6x}{6}$
$x=7$

6) $7x - 12 = 6 + 2x$

$-2x+12 \quad +12-2x$
$\frac{5x}{5} = \frac{18}{5}$
$x=\frac{18}{5}$ or $3\frac{3}{5}$

7) $3(4x + 2) - 2(5x - 3) = 12$

$12x+6-10x+6=12$
$2x+12=12$
$-12 \quad -12$
$2x=0$
$x=0$

8) $-2(4x - 5) = 8 - 6(2x + 1)$

$-8x+10=8-12x-6$
$-8x+10=2-12x$
$+12x \quad -10 \quad -10+12x$
$\frac{4x}{4} = \frac{-8}{4}$
$x=-2$

9) $3(x + 12) = 8 - 4x$

$3x+36=8-4x$
$+4x \quad -36 \quad -36+4x$
$\frac{7x}{7} = \frac{-28}{7}$
$x=-4$

10) $6(x - 10) = 5x - 42 - 2x$

$6x-60=3x-42$
$-3x \quad -3x$
$3x-60=-42$
$+60 \quad +60$
$3x=18$
$\frac{3x}{3}=\frac{18}{3}$
$x=6$

11) $2(4x + 2) = 4x - 12(x - 1)$

$8x+4=4x-12x+12$
$8x+4=-8x+12$
$+8x \quad -4 \quad +8x \quad -4$
$\frac{16x}{16} = \frac{8}{16}$
$x=\frac{1}{2}$

12) $\frac{2}{3}(3x + 9) = -2(2x + 6)$

$2x+6=-4x-12$
$+4x \quad -6 \quad +4x \quad -6$
$\frac{6x}{6} = \frac{-18}{6}$
$x=-3$